

# SLT-Guided Residual Seeds: Weakness, Evidence Geometry, and Refinement DAGs for Transformer Adaptation

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## Abstract

We formalize the connection between Singular Learning Theory (SLT) and the Hyperseed framework for modular transformer adaptation. Five claims are developed: (1) residual seeds are local evidence factors under a product-model assumption with an empirically testable interaction remainder; (2) LLC interaction information provides a proposed diagnostic taxonomy for seed coupling; (3) seeds participate in refinement DAGs scored by a seven-term pregate objective; (4) additivity deviation  $\Delta_{\text{add}}$  is a measurable associative proxy for compositional evidence leakage (the single novel bridge claim); and (5) the seed-ecology pattern extends to all LLC-estimable component-decomposable systems (restricted universality conjecture). A fully preregistered experimental protocol specifies the LLC estimator, multiplicity correction via Holm–Bonferroni, synthetic controls with pilot/confirmation split, and falsification criteria. All claims are framed with explicit epistemic caveats: conditionals rather than biconditionals, association rather than causation, and confidence intervals with family-wise error rate control.

## 1 Introduction

Singular Learning Theory [12] provides a Bayesian framework for understanding the geometry of loss landscapes in models whose parameter-to-function map is not injective. The *local learning coefficient* (LLC, also called the *real log canonical threshold*)  $\lambda$  controls the leading correction to the Bayesian free energy:

$$F_n = n \hat{L}_n + \lambda \log n - (m - 1) \log \log n + O(1), \quad (1)$$

where  $n$  is sample size,  $\hat{L}_n$  the empirical loss, and  $m$  the multiplicity of the dominant singularity.

The Hyperseed framework [1, 2] proposes that residual corrections (“seeds”) attached to a frozen base model form an *ecology*: they are promoted, demoted, split, merged, and refined through a directed acyclic graph of structural moves. The central question is: *what scoring criterion should govern this ecology?*

This note argues that SLT’s evidence geometry—specifically, the decomposition of the LLC across components and the deviations from additivity—provides the correct scoring criterion. We formalize five claims, each with explicit assumptions and epistemic caveats. We then specify a preregistered experimental protocol that makes the claims falsifiable.

**Scope.** This note imports definitions from the Causal Fibres framework (Note 0014) [9, 10] without re-deriving them. The novel contribution is Claim 4: the identification of  $\Delta_{\text{add}}$  as a quantitative associative bridge between SLT geometry and Hyperseed compositional structure.

## 2 Imports from Note 0014

The following definitions are imported from the Causal Fibres framework [9] and Note 0014 [10]. They are cited here for reference; see the original sources for derivations.

- (i) **Frozen interface** (Def. 2.1 of [9]): a fixed API boundary between a base model  $B$  and corrective components, through which information flows are defined and measurable.
- (ii) **Matched budget** (Def. 2.2 of [9]): a constraint ensuring that compared components receive equivalent computational resources during evaluation.
- (iii) **Evidence record** (Def. 2.3 of [9]): the accumulated Bayesian evidence for a component’s contribution, scored via log-marginal-likelihood ratios.
- (iv) **Counterfactual factor closure** (Def. 2.5 of [9]): the condition under which a factorization of the evidence is closed under counterfactual interventions on component boundaries.
- (v) **H0–H6 hypothesis ladder** (Def. 2.8 of [9]): a hierarchy of increasingly strong hypotheses about representation quality, from null (H0: no better than random) to strong compositional (H6: full counterfactual factor closure).

These imports provide the scaffold within which our SLT-based claims operate. Claims 1–5 add LLC-based scoring criteria to 0014’s structural framework; they do not redefine the framework itself.

## 3 Claim 1: Residual Seeds as Local Evidence Factors

### 3.1 Definitions

**Definition 3.1** (Residual Seed). Let  $B$  be a frozen base model with parameter space  $\Theta_B$ . A *residual seed* is a parameterized correction  $r_c : \mathcal{X} \rightarrow \mathcal{Y}$  with parameters  $\theta_c \in \Theta_c$ , such that the composite model computes  $f(x; \theta_B, \theta_c) = B(x) + r_c(x; \theta_c)$ . A *factorization* is a collection  $\mathcal{F} = \{r_1, \dots, r_k\}$  of seeds with associated parameter subspaces  $\Theta_{c_1}, \dots, \Theta_{c_k}$ .

**Definition 3.2** (Evidence Neighborhood). For a seed  $r_i$  in context  $c$  (dataset, task, or distribution), the *evidence neighborhood*  $\mathcal{W}(r_i, c)$  is the connected component of the set  $\{\theta_i \in \Theta_i : K_c(\theta_i) \leq \epsilon\}$  containing the optimal parameters, where  $K_c$  is the Kullback–Leibler divergence from the true distribution in context  $c$ .

**Definition 3.3** (Interaction Remainder). Given a factorization  $\mathcal{F} = \{r_1, \dots, r_k\}$  with joint parameter  $\theta = (\theta_1, \dots, \theta_k)$ , define the *interaction remainder*

$$R(\theta) = K(\theta) - \sum_{c=1}^k K_c(\theta_c), \quad (2)$$

where  $K(\theta)$  is the KL divergence of the joint model and  $K_c(\theta_c)$  is the KL divergence restricted to component  $c$ .

### 3.2 Assumptions

**Assumption 3.4** (Product-Model Assumption). The parameter space admits a product factorization  $\theta = (\theta_1, \dots, \theta_k)$  such that the prior factorizes:  $\pi(\theta) = \prod_{c=1}^k \pi_c(\theta_c)$ .

**Assumption 3.5** (Asymptotic Regime). The sample size  $n$  is sufficiently large relative to the singularity structure that the free-energy asymptotic (1) applies (Watanabe’s main theorem [12]).

**Assumption 3.6** (Dominated Interaction Hypothesis). There exists a threshold  $\eta > 0$  such that  $|R(\theta)| < \eta$  for all  $\theta$  in the evidence neighborhood of the joint model. **This is a testable hypothesis, not an axiom.**

### 3.3 Proposition

**Proposition 3.7** (LLC Additivity under Factorization). *Under Assumptions 3.4–3.6, the following hold:*

- (a) **LLC additivity:**  $\lambda_{\text{joint}} = \sum_{c=1}^k \lambda_c + O(\eta)$ .
- (b) **Evidence factorization:**  $\log Z_n(\theta) = \sum_{c=1}^k \log Z_n(\theta_c) + O(\eta \log n)$ .
- (c) **Quasi-independence:** Each seed  $r_i$  has an evidence neighborhood  $\mathcal{W}(r_i, c)$  that is approximately independent of other seeds’ evidence neighborhoods, with coupling bounded by  $O(\eta)$ .

*Epistemic Caveat 3.8* (Limits of the Factorization). The smallness of  $R$  is a *testable hypothesis*, not a sufficient proof of meaningful factorization. Even if  $R$  is empirically small for a given model, this does not establish that the decomposition is coordinate-invariant or structure-revealing—it establishes only that the product-prior decomposition is a good approximation at the observed sample size. The decomposition could break under distribution shift, scale change, or different priors.

The implication is conditional: factorization  $\Rightarrow$  LLC additivity (under the assumptions). The converse does not hold in general; LLC additivity could arise from cancellations in the interaction remainder.

*Remark 3.9* (Relation to Source Claims). The evidence-factorization result builds on “If the evidence factorizes across modules, the global LLC is the sum of module LLCs” [1]. The promotion criterion derives from “Do not promote a residual basis because it reconstructs well. Promote it because  $R$  is small” [2]. The evidence-ratio to soft-accuracy bridge with  $O(\lambda/n)$  correction follows [3].

*Remark 3.10* (Non-Overlap with 0014). Note 0014 defines frozen interfaces, teacher-gap closure, and representation contests. Claim 1 adds the LLC-additivity criterion as the promotion gate *within* 0014’s frozen-interface scaffold. It does not redefine the interface.

## 4 Claim 2: LLC Interaction Information as a Proposed Diagnostic

**Definition 4.1** (LLC Interaction Information). For seeds  $a, b$  in a factorization  $\mathcal{F}$ , the *LLC interaction information* is

$$I_\lambda(a; b) = \lambda_a + \lambda_b - \lambda_{ab}, \quad (3)$$

where  $\lambda_a, \lambda_b$  are the per-seed LLCs and  $\lambda_{ab}$  is the LLC of the joint  $(a, b)$  subsystem.

**Definition 4.2** (Surrogate Estimators). Two computationally tractable surrogates for  $I_\lambda$ :

- (a) **Mixed-Hessian norm:**  $\|H_{ab}\| = \|\nabla_{\theta_a} \nabla_{\theta_b} L(\theta)\|_F$ , evaluated at the MAP or posterior mean.
- (b) **Commutator dominance:**  $\text{Comm}(a, b) = \|[J_a, J_b]\|_F$ , where  $J_a, J_b$  are the Jacobians of the component maps  $r_a, r_b$  with respect to their parameters.

**Proposed Diagnostic Taxonomy.** The following interpretation of  $I_\lambda$  is a *proposed diagnostic framework*, not a proven classifier:

| Sign of $I_\lambda$   | Proposed Interpretation                               | Status                  |
|-----------------------|-------------------------------------------------------|-------------------------|
| $I_\lambda > 0$       | Redundant/shared singular structure (merge candidate) | Hypothesis              |
| $I_\lambda < 0$       | Synergistic/emergent structure (assembly needed)      | Hypothesis              |
| $I_\lambda \approx 0$ | Factorizable independence                             | Consistent with Claim 1 |

*Epistemic Caveat 4.3* (Surrogates Are Not Interchangeable with  $I_\lambda$ ). The surrogates  $\|H_{ab}\|$  and  $\text{Comm}(a, b)$  correlate with the true LLC interaction information under regularity conditions, but they measure different geometric quantities: local curvature cross-terms vs. singularity-structure overlap. Their agreement with  $I_\lambda$  degrades near degenerate singularities. Discrepancies between surrogates should be reported as evidence of estimator limitations, not resolved by fiat [1, 2].

**Definition 4.4** (Surrogate Agreement). Define the surrogate agreement score

$$\delta(\text{Comm}, \|H_{ab}\|, I_\lambda) = \max(|\text{Comm} - I_\lambda|, \|\|H_{ab}\| - I_\lambda\|), \quad (4)$$

after appropriate normalization. Large  $\delta$  indicates that the surrogates are unreliable for the pair  $(a, b)$ .

**Falsifiable Prediction.** In a small transformer with known modular structure (e.g., a two-task mixture), per-seed LLC estimates should yield  $I_\lambda \approx 0$  between task-specific seeds and  $I_\lambda > 0$  between capacity-sharing seeds [4]. Additionally, the surrogate measures should agree with direct LLC estimates to within preregistered tolerances. Surrogate–direct discrepancies, if found, would weaken the diagnostic taxonomy rather than the underlying SLT theory.

## 5 Claim 3: Refinement DAGs as Operational Structure

Seeds do not exist in a fixed decomposition. They participate in a *refinement DAG* whose vertices are factorizations and whose edges are structural moves.

**Definition 5.1** (Refinement Move). A *refinement move*  $\mu : \mathcal{F} \rightarrow \mathcal{F}'$  transforms a factorization  $\mathcal{F}$  into a successor  $\mathcal{F}'$  via one of the following operations: SPLIT (one seed  $\rightarrow$  two), MERGE (two seeds  $\rightarrow$  one), MEDIATE (insert an intermediary), SPECIALIZE (restrict a seed’s domain), DEMOTE (remove a seed), or TRANSFER (reassign a seed to a different context).

**Definition 5.2** (Pregate Score). The *pregate score* of a factorization  $\mathcal{F}$  in context  $c$  is the seven-term objective [2]:

$$J(\mathcal{F}) = n \hat{L}_{\mathcal{F}} + \hat{\lambda}_{\mathcal{F}} \log n - \log \pi(\mathcal{F}) + \alpha \text{Comm}(\mathcal{F}) + \beta \text{Leak}(\mathcal{F}) + \gamma \text{Regret}(\mathcal{F}) + \delta \text{Interaction}(\mathcal{F}), \quad (5)$$

where  $\hat{L}_{\mathcal{F}}$  is the empirical loss under factorization  $\mathcal{F}$ ,  $\hat{\lambda}_{\mathcal{F}}$  the estimated LLC,  $\pi(\mathcal{F})$  the structural prior over factorizations, and the remaining terms penalize commutator dominance, evidence leakage, counterfactual regret, and interaction remainder, respectively. The coefficients  $\alpha, \beta, \gamma, \delta \geq 0$  are hyperparameters.

**Definition 5.3** (Associative Signature and Regime Signature). The *associative signature* of seed  $r_i$  in context  $c$  is the vector

$$\sigma(r_i, c) = (\lambda_i, \text{Comm}(r_i, \cdot), \text{Leak}(r_i), I_\lambda(r_i, \cdot)) \in \mathbb{R}^{d_\sigma}. \quad (6)$$

The *regime signature* of a factorization  $\mathcal{F}$  at time  $t$  is

$$\Sigma(\mathcal{F}, t) = (\sigma(r_1, c_t), \dots, \sigma(r_k, c_t)). \quad (7)$$

Abrupt shifts in  $\Sigma$  trigger refinement moves [5].

**Connection to 0014.** Note 0014’s H1 representation contest (Def. 2.8 of [9]) compares arm classes at matched budget. Claim 3 extends this to a *dynamic* contest: the contest is not a one-shot comparison but a DAG of refinement moves, each subject to the same evidence gates 0014 defines.

## 6 Claim 4: Additivity Deviation as Evidence-Leakage Proxy

**This is the single novel bridge claim.** SLT’s RLCT decomposition across components predicts that deviations from LLC additivity signal geometric interactions between singular regions [11]. In Hyperseed terms, these deviations are *associative signatures* of evidence leakage between refinement stages.

**Definition 6.1** (Additivity Deviation). For a factorization  $\mathcal{F} = \{c_1, \dots, c_k\}$ , the *additivity deviation* is

$$\Delta_{\text{add}}(\mathcal{F}) = \left| \lambda_{\text{joint}} - \sum_{c=1}^k \lambda_c \right|. \quad (8)$$

### 6.1 Epistemic Downgrades

*Epistemic Caveat 6.2* (No Biconditional).  $\Delta_{\text{add}} > 0$  is *necessary but not sufficient* evidence that factorization  $\mathcal{F}$  fails to decompose the evidence geometry. Finite-sample bias in LLC estimation, prior misspecification, and MCMC mixing artifacts can all produce non-zero  $\Delta_{\text{add}}$  even when the true singularity structure decomposes.

The claim is:  $\Delta_{\text{add}}$  *significantly exceeding zero, after accounting for estimator uncertainty, is evidence against clean factorization*. The converse ( $\Delta_{\text{add}} \approx 0$  implies good factorization) is better supported but still conditional on the product-prior assumption (Assumption 3.4).

*Epistemic Caveat 6.3* (Association, Not Causation). The prediction is that  $\Delta_{\text{add}}$  *correlates with* graph-theoretic cross-stage dependency in the refinement DAG. This is an associative claim. Upgrading to causal language requires an intervention that varies cross-stage coupling while **preserving parameter count and component decomposition**. Three candidate interventions, ordered by preference:

- (a) **Coupling-strength interpolation.** Parameterize cross-stage connections with a scalar  $\alpha \in [0, 1]$ . At  $\alpha = 1$  the architecture is unchanged; at  $\alpha = 0$  cross-stage information flow is severed. All parameters remain in the model (zeroed weights are frozen, not removed). Predict:  $\Delta_{\text{add}}$  decreases monotonically as  $\alpha \rightarrow 0$ .
- (b) **Calibrated noise injection** (secondary/optional). Add i.i.d. Gaussian noise  $\varepsilon \sim \mathcal{N}(0, \sigma^2)$  at inter-stage boundaries, sweeping  $\sigma$ . **Matching requirements:** at each  $\sigma$ , marginal activation variance and predictive loss must match the  $\alpha$ -sweep baseline within 5%. Without these calibrations, noise injection is a confound.
- (c) **Matched-complexity control** (if architectural changes are unavoidable). Pair each structural ablation with a control: the same architecture but re-initialized (random) weights, matching parameter count and decomposition.

Until an intervention is run, “ $\Delta_{\text{add}}$  tracks cross-stage dependency” means association.

### 6.2 Confidence-Interval Requirement

All reported  $\Delta_{\text{add}}$  values must include:

- Bootstrap or MCMC-derived confidence intervals on individual  $\lambda_c$  estimates.

- Propagated uncertainty on  $\Delta_{\text{add}}$  itself (via the delta method or bootstrap on the sum).
- Multiplicity-corrected significance:  $\Delta_{\text{add}}$  is declared “meaningfully non-zero” iff its Holm-adjusted  $p$ -value  $< 0.05$  across the full comparison family (see §8.2).

### 6.3 Falsifiable Experimental Check

Take a small transformer (or synthetic model with known singular structure). Compute per-layer LLC estimates via the `devinterp` local learning coefficient estimator with stated hyperparameters. Report  $\Delta_{\text{add}}$  with confidence intervals. Independently compute a graph-theoretic cross-stage dependency score on the refinement DAG. Test for statistically significant correlation (Spearman  $\rho$ , Holm-adjusted  $p < 0.05$ ).

- If  $\Delta_{\text{add}}$  is significantly elevated but the DAG shows clean stage separation, the associative claim is **falsified**.
- If they correlate, the Hyperseed–SLT bridge has non-trivial empirical support—but *not* causal evidence unless an intervention experiment (Caveat 6.3) is also run.

**Why Novel.** The individual pieces (LLC decomposition [1, 4], refinement DAGs [1], counterfactual factor closure [9]) exist in the source papers. The novel claim is that  $\Delta_{\text{add}}$  is the *quantitative observable* that connects SLT geometry to Hyperseed compositional structure—as a measurable, falsifiable associative bridge.

## 7 Claim 5: Restricted Adaptation Conjecture

The seed-ecology pattern—base process + controlled corrective factors, scored by evidence geometry, refined through a DAG of structural moves, monitored for regime change—extends beyond residual layers to a defined class of systems.

**Definition 7.1** (LLC-Estimable Component-Decomposable System). A system belongs to the *LLC-estimable component-decomposable* class iff:

- (C1) It admits a finite parametric description  $\theta \in \Theta \subset \mathbb{R}^d$ .
- (C2) Its loss function  $L(\theta)$  is real-analytic (or admits a resolution of singularities).
- (C3) Local learning coefficients are estimable at each component (the posterior concentrates sufficiently for MCMC-based LLC estimation to converge).
- (C4) It has an explicit component decomposition  $\mathcal{F} = \{c_1, \dots, c_k\}$  with defined parameter subspaces  $\Theta_{c_i}$  and a product-prior factorization.
- (C5) The interaction remainder  $R(\theta)$  from Definition 3.3 is well-defined and estimable.

**Conjecture 7.2** (Restricted Hyperseed Universality). *Any system satisfying conditions (C1)–(C5) that maintains:*

- (i) *a base predictive process,*
- (ii) *an ecology of modular corrective seeds scored by local evidence (LLC-additivity checks, not reconstruction error alone), and*
- (iii) *a refinement DAG over candidate factorizations*

*will exhibit the same LLC-additivity / interaction-information / regime-signature dynamics described in Claims 1–4.*

### Conjectured Properties within this Class.

- **Incremental compression:** corrective components are compression features for the base model’s error stream.
- **Evidence gating:** promotion/demotion is governed by LLC-additivity checks, not reconstruction error alone.
- **Regime detection:** LLC/signature shifts trigger structural revision [5].
- **Interaction diagnostics:** coupling type is diagnosed via  $I_\lambda$  (Claim 2’s proposed taxonomy).
- **Goal-stability limits:** SLT diagnostics detect structural shifts but not semantic truth; the semantics route via distinctions  $\rightarrow$  partitions  $\rightarrow$  symmetries  $\rightarrow$  singularities adds the missing layer [6, 7].

*Epistemic Caveat 7.3* (Scope Exclusions). Explicitly excluded from Conjecture 7.2: non-parametric models, models where LLC estimation does not converge, systems without a natural component decomposition, and infinite-dimensional parameter spaces (unless a finite-dimensional effective description exists).

## 8 Preregistered Experimental Protocol

Before running the falsification experiment for Claims 1, 2, and 4, the following must be pre-registered.

### 8.1 LLC Estimator Specification

**Protocol 8.1** (LLC Estimator).

- **Estimator:** devinterp local learning coefficient estimator (Lau et al., 2024).
- **MCMC method:** Stochastic Gradient Langevin Dynamics (SGLD) with specified step-size schedule, number of chains, burn-in length, and thinning interval. All hyperparameters are fixed before data collection.
- **Convergence diagnostic:**  $\hat{R} < 1.05$  across chains; effective sample size  $\geq 200$  per component.
- **Hyperparameters to freeze:** learning rate, temperature schedule, number of posterior samples, number of chains, burn-in duration, and thinning interval.

### 8.2 Uncertainty Quantification and Multiplicity Correction

**Per-component uncertainty.**

- **Per-component  $\lambda_c$ :** Bootstrap confidence intervals ( $B \geq 1000$  resamples) or MCMC posterior credible intervals (95%).
- $\Delta_{\text{add}}$ : Propagated via the delta method or bootstrap on the sum; reported with 95% CI.
- $I_\lambda(a, b)$ : Same uncertainty propagation; reported with 95% CI.
- **Unadjusted significance threshold:**  $\Delta_{\text{add}}$  is declared “meaningfully non-zero” iff the 95% CI excludes zero.

**Comparison family.** The experiment involves simultaneous statistical tests. Define the comparison family as:

- **$\Delta_{\text{add}}$  tests:**  $k$  coupling levels  $\times m$  components =  $k \cdot m$  comparisons.
- **$I_\lambda$  tests:**  $\binom{m}{2} = m(m-1)/2$  pairwise interaction-information comparisons.
- **Total family size:**

$$N = k \cdot m + \frac{m(m-1)}{2}. \quad (9)$$

With  $k = 11$  ( $\alpha$  from 0.0 to 1.0 in steps of 0.1) and  $m$  components (fixed before data collection),  $N$  is stated explicitly in the preregistration.

**Holm–Bonferroni Procedure.** The Holm–Bonferroni sequential rejection procedure (Holm, 1979) controls the family-wise error rate (FWER) at  $\alpha_{\text{family}} = 0.05$ :

1. Order all  $N$   $p$ -values:  $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(N)}$ .
2. Reject  $H_{0,(j)}$  iff  $p_{(j)} \leq \alpha_{\text{family}}/(N - j + 1)$  for all  $j' \leq j$ .
3. Equivalently, report Holm-adjusted  $p$ -values:

$$\tilde{p}_{(j)} = \max_{j' \leq j} \{(N - j' + 1) \cdot p_{(j')}\}. \quad (10)$$

4. An individual comparison is declared significant iff its Holm-adjusted  $p$ -value  $< 0.05$ .

**Adjusted Confidence Intervals.** Where CIs are reported alongside Holm-corrected tests, use Bonferroni-adjusted CIs at level  $1 - \alpha_{\text{family}}/N$  per comparison (conservative but consistent with FWER control). These replace unadjusted 95% CIs in all multi-comparison contexts.

**Rationale for Holm over Alternatives.** Holm–Bonferroni controls FWER (appropriate here because a single false positive about coupling structure would undermine the bridge claim) while being strictly more powerful than Bonferroni. FDR-controlling methods (Benjamini–Hochberg) would be appropriate for screening many components for follow-up, but here each comparison bears directly on the claim.

### 8.3 Component-to-DAG Mapping

**Protocol 8.2** (Component-to-DAG Mapping).

- **Component definition:** Each residual block (or attention head / MLP sublayer, depending on granularity—to be fixed before experiment) is one component  $c_i$ .
- **DAG construction:** Edges from  $c_i$  to  $c_j$  iff information flow exists (measured by gradient attribution or activation patching). Edge weights from mutual information or ablation impact.
- **Cross-stage dependency score:** Weighted edge density between non-adjacent DAG stages. The formal graph-theoretic definition is stated before data collection.



## 8.4 Synthetic Controls with Pilot/Confirmation Split

**Split Design.** Synthetic control data is generated once and split into two disjoint sets before any analysis:

- **Pilot calibration set (50%):** Used *exactly once* for threshold calibration (§8.6). After thresholds are frozen, this data is permanently set aside.
- **Held-out confirmation set (50%):** Reserved exclusively for admission tests. Never examined during calibration.
- **Contingency:** If the pilot set is too small for stable threshold estimation (bootstrap CV of threshold estimates  $> 0.2$ ), the split shifts to 60/40 pilot/confirmation. This contingency is preregistered.

**Null/Negative Control.** Two independent MLPs sharing no parameters, trained on disjoint tasks from the same data distribution. Components are a priori independent by construction.

- **Expected:**  $\Delta_{\text{add}} \approx 0$ ,  $I_\lambda \approx 0$  between the two networks,  $R < \eta$ .
- **Gate:** If the estimator reports non-zero  $\Delta_{\text{add}}$  on this null control, the estimator or its hyperparameters are miscalibrated; do not proceed.

**Positive Control.** Two MLPs ( $\text{component}_1$ ,  $\text{component}_2$ ) connected by a fixed linear coupling layer  $W_{\text{mix}}$  of known rank  $r$ , trained jointly on the same task mixture. The component decomposition is defined a priori:  $\text{component}_1 = \text{MLP}_1$  parameters,  $\text{component}_2 = \text{MLP}_2$  parameters, coupling =  $W_{\text{mix}}$ .

*Epistemic Caveat 8.3 (No Analytic Proportionality).* The prediction is:  $\Delta_{\text{add}}$  *is expected to be significantly nonzero (Holm-adjusted  $p < 0.05$ ) on the positive control, with the effect size estimated from the pilot calibration set.* No analytic proportionality between  $\Delta_{\text{add}}$  and  $\text{rank}(W_{\text{mix}}) \cdot \|W_{\text{mix}}\|_2$  is claimed. Computing the actual RLCT contribution of the coupling requires a resolution of singularities for the specific loss surface, which has not been performed. If a tighter analytic bound becomes available (e.g., via explicit resolution of the  $W_{\text{mix}}$  singularity under a Gaussian prior), it will be stated as a separate, independently testable prediction.

**Decision Rule (Confirmation Set Only).** Both controls must pass before claims about the test model are made:

- Null: Holm-adjusted  $p > 0.05$  for  $\Delta_{\text{add}}$ .
- Positive: Holm-adjusted  $p < 0.05$  for  $\Delta_{\text{add}}$ .

## 8.5 Model and Data

- **Test model:** Small transformer ( $\leq 6$  layers) trained on a two-task mixture.
- **Held-out model:** A second, architecturally distinct model (e.g., a 4-layer transformer if the test model is 6 layers, or a convolutional residual network) trained on the same data. Predictions for this held-out model are registered before fitting. If Claims 1–4 hold on the test model but fail on the held-out model, the claims are model-specific, not general.
- **Training data:** Fixed dataset, split specified before training.
- **LLC estimation data:** Training split (for loss-landscape exploration).
- **Evaluation data:** Held-out split, not used for LLC estimation or estimator tuning.

## 8.6 Numerical Decision Thresholds

All significance tests across the  $\alpha$ -coupling sweep and component comparisons are corrected for multiplicity via the Holm–Bonferroni procedure (§8.2). Thresholds are calibrated on the pilot set (§8.4) and frozen before confirmation data is examined.

- **$\Delta_{\text{add}}$  significance:**  $\Delta_{\text{add}}$  is declared “meaningfully non-zero” iff its Holm-adjusted  $p$ -value  $< 0.05$ . Reported with Bonferroni-adjusted CI at level  $1 - 0.05/N$ .
- **Interaction remainder threshold:**  $\eta = 0.1 \times \lambda_{\text{max}}$ , where  $\lambda_{\text{max}}$  is the largest per-component LLC.  $R/\lambda_{\text{max}} < 0.1$  counts as “dominated.”
- **$I_\lambda$  significance:**  $|I_\lambda(a, b)|$  is declared non-negligible iff its Holm-adjusted  $p$ -value  $< 0.05$ .
- **Correlation threshold (Claim 4 primary test):** Spearman  $\rho$  with Holm-adjusted  $p < 0.05$  (one-tailed). The minimum detectable  $\rho$  is set using the pilot calibration set, targeting  $\geq 80\%$  power at the pilot-observed effect size. If the pilot effect is too small for 80% power, report achieved power and note the study is underpowered.
- **Sign accuracy (Claim 2):** Agreement between  $\text{sign}(I_\lambda)$  and known modular structure. Threshold set at  $2\sigma$  above null expectation (random assignment baseline), calibrated on pilot data. Cohen’s  $\kappa$  threshold frozen before confirmation.
- **Factorization prevalence (Claim 1):**  $R < \eta$  for a preregistered fraction of components. Threshold set at the ROC-optimal discrimination point from pilot data. If controls show perfect separation, use 80% (conservative).

**Circularity Prevention.** The three calibration steps each consume the pilot set. Once thresholds are frozen, the pilot data is permanently set aside. All subsequent control-pass/fail decisions and claim evaluations use only the held-out confirmation set. No threshold is revised after examining confirmation data.

## 8.7 Analysis Plan

- **Primary test (Claim 4):** Spearman correlation between per-component  $\Delta_{\text{add}}$  and cross-stage dependency score. Report coefficient,  $p$ -value, and CI.
- **Secondary test (Claim 2):** Agreement between  $\text{sign}(I_\lambda)$  and known modular structure. Report accuracy, Cohen’s  $\kappa$ .
- **Tertiary test (Claim 1):** Fraction of components where  $R < \eta$  ( $\eta$  preregistered). Report proportion and bootstrap CI.

### Falsification Criteria.

- Claim 4 is falsified if the Holm-adjusted  $p$ -value for the  $\Delta_{\text{add}}$ –dependency correlation exceeds 0.05.
- Claim 2’s taxonomy is weakened if sign accuracy falls below the pilot-calibrated threshold.
- Claim 1’s factorization hypothesis is weakened if the proportion of components with  $R < \eta$  falls below the pilot-calibrated prevalence threshold.
- All thresholds are those frozen after pilot calibration (§8.6); none are adjusted post hoc.

## Intervention Design (Eventual Causal Upgrade of Claim 4).

- **Primary:** Coupling-strength interpolation ( $\alpha$  sweep from 1.0 to 0.0 in steps of 0.1) at selected stage boundaries, preserving parameter count and decomposition. Re-estimate  $\Delta_{\text{add}}$  at each setting; test for monotonic decrease. Holm–Bonferroni applied across all  $\alpha$  levels and components jointly.
- **Secondary (optional):** Calibrated noise injection ( $\sigma$  sweep) with mandatory marginal-activation-variance and predictive-loss matching (Caveat 6.3(b)). If architectural ablation is used, include a matched-complexity control.
- This is planned as a follow-up, not part of the initial preregistration.

## 9 Open Questions

*Open Problem 1* (Coordinate Invariance). The product-prior decomposition (Assumption 3.4) is coordinate-dependent. Under what conditions does the LLC-additivity result (Proposition 3.7) hold in reparameterized coordinates? Is there a coordinate-free characterization of “good factorization”?

*Open Problem 2* (Resolution of Singularities for Coupled Components). The positive control (§8.4) uses a coupling matrix  $W_{\text{mix}}$ . What is the RLCT of the joint model as a function of  $\text{rank}(W_{\text{mix}})$  and the component RLCTs? An explicit resolution would convert Claim 4’s associative prediction into a quantitative one.

*Open Problem 3* (Scaling of the Interaction Remainder). How does  $R(\theta)$  scale with model depth, width, and dataset size? If  $R$  grows faster than  $\lambda_{\text{max}}$ , the “dominated interaction” hypothesis (Assumption 3.6) fails at scale, and the factorization framework may not apply to large models.

*Open Problem 4* (Semantic Registration). Claim 5 notes that SLT diagnostics detect structural shifts but not semantic truth [6]. How should the distinctions  $\rightarrow$  partitions  $\rightarrow$  symmetries  $\rightarrow$  singularities pipeline [7] be formalized to provide the missing semantic layer?

*Open Problem 5* (Beyond Real-Analytic Loss). Condition (C2) of Definition 7.1 restricts to real-analytic loss functions. Many practical losses (e.g., those involving ReLU activations) are only piecewise analytic. Can the framework be extended via stratified resolution of singularities, or does the LLC decomposition fundamentally require analyticity?

*Open Problem 6* (Dynamic Pregate Coefficients). The pregate score (5) has fixed coefficients  $\alpha, \beta, \gamma, \delta$ . Should these be adapted over the refinement DAG (e.g., via meta-learning on the DAG trajectory)? What are the convergence implications of a non-stationary scoring function?

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